MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 5**

The derivative as a function and the differentiation rules (Sections 2.2–2.4, Lectures 8–9); the chain rule and implicit differentiation (Sections 2.5–2.6, Lecture 10).

Key facts to have at hand (Lectures 8–10)

- **The derivative as a function:** $f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, defined at every x for which the limit exists. The domain of f' may be *smaller* than the domain of f. Other notations: y', $\dfrac{dy}{dx}$, $\dfrac{d}{dx}f(x)$. Then $f'' = (f')'$, f''', $f^{(4)}$, …; if $s(t)$ is position, s' is velocity and s'' is acceleration.
- **Reading f' off the graph of f:** rising $\Rightarrow f' > 0$; falling $\Rightarrow f' < 0$; horizontal tangent $\Rightarrow f' = 0$; the steeper the curve, the further f' is from the axis.
- **Differentiable $\Rightarrow$ continuous** — never the other way round. Three ways to have no derivative at a: a *corner* (the one-sided slopes exist and differ), a *vertical tangent* (the difference quotient runs to $\pm\infty$), a *discontinuity*.
- **The rules:** $(c)' = 0$; $(x^n)' = nx^{n-1}$ for every real n; $(cf)' = cf'$; $(f \pm g)' = f' \pm g'$; $(fg)' = f\,g' + g\,f'$; $\left(\dfrac{f}{g}\right)' = \dfrac{g\,f' - f\,g'}{g^2}$ $(g \neq 0)$. *Rewrite before you reach for a rule:* $\sqrt{x} = x^{1/2}$, $\dfrac{5}{x^4} = 5x^{-4}$, $\dfrac{x^3+1}{x} = x^2 + \dfrac{1}{x}$.
- **Trigonometric derivatives** (radians, always): $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$, $(\cot x)' = -\csc^2 x$, $(\sec x)' = \sec x \tan x$, $(\csc x)' = -\csc x \cot x$. The three "co" functions carry the minus sign. Behind them: $\lim\limits_{h\to 0} \dfrac{\sin h}{h} = 1$ and $\lim\limits_{h\to 0} \dfrac{\cos h - 1}{h} = 0$.
- **The chain rule:** if $y = f(u)$ and $u = g(x)$, then $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$, i.e. $\big[f(g(x))\big]' = f'\big(g(x)\big) \cdot g'(x)$: *outside first, inside left alone, then times the inside's derivative* — and f' is fed $g(x)$, not x. For a chain of three, three factors. Special cases: $\big[g(x)^n\big]' = n\,g(x)^{n-1} g'(x)$, $\big[\sin g(x)\big]' = \cos g(x) \cdot g'(x)$.
- **Implicit differentiation:** to find y' from an equation in x and y, differentiate both sides with respect to x — *every y you differentiate brings a y'* (chain rule: $(y^3)' = 3y^2 y'$, $(\sin y)' = \cos y \cdot y'$, $(xy)' = y + xy'$) — then solve for y'. The answer may contain x and y; substitute the point *after* solving. For y'', differentiate y' again and replace y' by what you found. The method assumes the curve is a graph near the point; where it is not (the circle $x^2 + y^2 = 169$ at $(13, 0)$), no y' solves the equation.

Part I · The derivative as a function; the rules; the trigonometric derivatives (Sections 2.2–2.4) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is differentiable at a, then f is continuous at a.
2. If f is continuous at a, then f is differentiable at a.
3. If the graph of f has a corner at a, then f is not continuous at a.
4. Wherever the graph of f is below the x-axis, the graph of f' is below the x-axis.
5. The derivative of $\dfrac{1}{x^2}$ is $\dfrac{1}{2x}$.
6. $\dfrac{d}{dx}\big(x \cdot x^2\big) = 1 \cdot 2x = 2x$.
7. $\dfrac{d}{dx}\cos x = \sin x$.
8. The derivative of a polynomial of degree 5 is a polynomial of degree 4.
9. If $f(x) = \pi^3$, then $f'(x) = 3\pi^2$.
10. If $s(t)$ is the position of a particle at time t, then $s''(t)$ is its acceleration.

Problem 1 *The rules — and the rewriting that comes before them.*

Differentiate. Rewrite roots, reciprocals and quotients whose denominator is a single power of x before you reach for a rule, and name the rule you use.

(a) $f(x) = 2x^5 - 4x^3 + 7x - 9$

(b) $g(x) = 8\sqrt{x} - \dfrac{3}{x} + 6\sqrt[3]{x}$

(c) $F(t) = (t^2 + 3)(2t - 5)$ (two ways: multiply out first; by the product rule)

(d) $y = \dfrac{x^3 - 4x + 6}{x^2}$ (without the quotient rule)

(e) $y = \dfrac{2x + 3}{x^2 + 1}$

(f) $G(x) = (x - 1)(x + 1)(x^2 + 1)$

(g) For $P(x) = x^4 - 6x^2 + 2x$ find P', P'', P''', $P^{(4)}$ and $P^{(5)}$.

Problem 2 *Product and quotient from a table.*

Two functions f and g are differentiable, and

$$f(1) = 3, \quad f'(1) = -2, \quad g(1) = 4, \quad g'(1) = 5; \qquad f(2) = -1, \quad f'(2) = 6, \quad g(2) = 2, \quad g'(2) = -3.$$

(a) $(fg)'(1)$

(b) $\left(\dfrac{f}{g}\right)'(1)$

(c) $\left(\dfrac{g}{f}\right)'(1)$

(d) $\dfrac{d}{dx}\big[x^2 f(x)\big]$ at $x = 2$

(e) $\dfrac{d}{dx}\left[\dfrac{f(x) + g(x)}{x}\right]$ at $x = 2$

(f) The equation of the tangent line to $y = f(x)g(x)$ at $x = 2$.

Problem 3 *Trigonometric derivatives.*

(a) $y = x^3 \cos x$

(b) $y = \dfrac{\sin x + \cos x}{\sin x - \cos x}$ (simplify the answer)

(c) $y = 4\tan x - 3\sec x$

(d) $y = x\cos x - \sin x$ (simplify the answer)

(e) Show that $y = 4\sin x + 5\cos x$ satisfies $y'' + y = 0$.

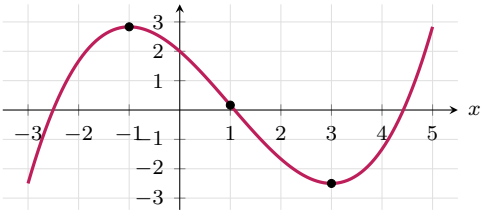
(f) Find all points of the graph of $y = \sin x + \sqrt{3}\cos x$, $0 \le x \le 2\pi$, at which the tangent line is horizontal.

(g) (Two of the four formulas that Lecture 9 left as a homework exercise.) Starting from $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$, derive the formulas for $(\sec x)'$ and $(\cot x)'$ by the quotient rule.

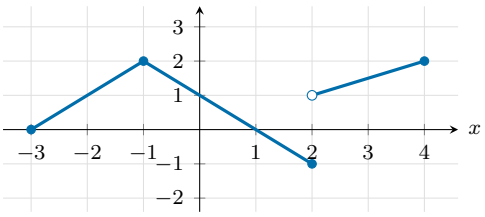
Problem 4 *Reading f' off the graph of f; where there is no derivative.*

The graphs of two functions f and g are shown: f is defined for every x (only the part over $[-3,5]$ is drawn), and g is defined on $[-3,4]$ only. (Blank axes for your sketches are below.)

$y = f(x)$



$y = g(x)$



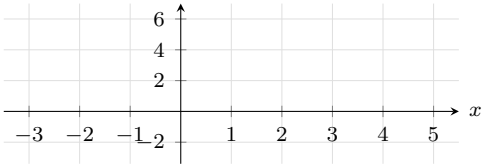
(a) For f: on which intervals is $f' > 0$, on which $f' < 0$, and where is $f' = 0$? At which x is f' smallest (the descent steepest)? Sketch the graph of f'.

(b) For g ($g(2) = -1$, and the open dot is not on the graph): sketch the graph of g', with its domain.

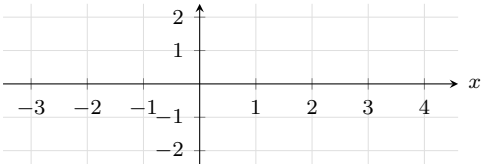
(c) At which numbers in $(-3,4)$ is g not differentiable? Give the reason for each, and for the corner compute the two one-sided limits of the difference quotient.

(d) Show from the definition that $k(x) = \sqrt[3]{x-1}$ has a vertical tangent at $x = 1$: compute the limit $\lim_{h\to 0} \dfrac{k(1+h)-k(1)}{h}$ and say what it tells you.

$y = f'(x)$ — your sketch



$y = g'(x)$ — your sketch



Problem 5 *Differentiable, or only continuous?*

(a) Find the constants a and b for which $f(x) = \begin{cases} x^2 - 2x, & x \le 3, \\ ax + b, & x > 3 \end{cases}$ is differentiable at every number.

(b) For which c is $g(x) = \begin{cases} cx^2, & x \le 1, \\ 3x - 1, & x > 1 \end{cases}$ continuous at 1? For that c, is g differentiable at 1? (Compare the one-sided slopes.)

(c) Show from the definition that $p(x) = x|x|$ is differentiable at 0, and find $p'(0)$. (Then write a formula for $p'(x)$ for every x.)

(d) At which numbers is $q(x) = |x^2 - 4x|$ not differentiable? Give the one-sided slopes there.

Problem 6 *The derivative as a function, from the definition.*

Find $f'(x)$ from the definition, $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, keeping "lim" in front of every line until the limit is taken. State the domain of f and the domain of f'.

(a) $f(x) = 3x^2 - x + 4$

(b) $f(x) = \dfrac{1}{x+3}$

(c) $f(x) = \sqrt{x-2}$

(d) $f(x) = x^3 - 4x$; then find $f''(x)$ the same way.

Part II · The chain rule and implicit differentiation (Sections 2.5–2.6) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. $\dfrac{d}{dx}\sin(x^2) = \cos(x^2)$.
2. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$.
3. $\dfrac{d}{dx}(x^2+1)^{10} = 10(x^2+1)^9$.
4. $\dfrac{d}{dx}\sqrt{x^2+1} = \dfrac{x}{\sqrt{x^2+1}}$.
5. If $f'(2) = 3$, $g(1) = 2$ and $g'(1) = 5$, then $(f \circ g)'(1) = 15$.
6. $(f \circ g)'(x) = f'(x)\,g'(x)$ for all differentiable f and g.
7. On the circle $x^2 + y^2 = 25$ the tangent line at $(3, 4)$ has slope $-\dfrac{3}{4}$.
8. When y is a function of x, $\dfrac{d}{dx}\big(y^3\big) = 3y^2$.
9. When y is a function of x, $\dfrac{d}{dx}\big(\sin y\big) = \cos y \cdot y'$.
10. When y is a function of x, $\dfrac{d}{dx}\big(xy\big) = x\,y'$.

Problem 7 *The chain rule.*

Differentiate. For each function say what the *outside* is and what the *inside* is before you write anything.

(a) $y = (3x^2 - 5)^4$

(b) $y = \sqrt{5x^2 + 1}$

(c) $y = \dfrac{1}{(x^2 + 2x)^3}$

(d) $y = \tan(x^3)$

(e) $y = x^2(1 - 2x)^5$ (simplify: take out the common factor)

(f) $y = \left(\dfrac{x-1}{x+2}\right)^3$

Problem 8 *Chains of three.*

Peel from the outside in; every layer contributes one factor. (Parts (a) and (b) are the two that Lecture 10's chain-of-three slide set for the tutorial.)

(a) $y = \sqrt{1 + \sin 3x}$

(b) $y = \cos^2(5x)$ (simplify with a double-angle formula)

(c) $y = (1 + \tan 2x)^3$

(d) $y = \cos\Big(\sqrt{1 + x^2}\Big)$

Problem 9 *The chain rule from a table.*

f and g are differentiable, and

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-2	2	4
2	1	5	3	-1
3	2	6	1	3

Find: (a) $(f\circ g)'(1)$; (b) $(g\circ f)'(3)$; (c) $\dfrac{d}{dx}\big[f(x)\big]^3$ at $x=1$; (d) $\dfrac{d}{dx}f(2x-1)$ at $x=2$; (e) $\dfrac{d}{dx}\sqrt{g(x)}$ at $x=3$; (f) $(f\circ f)'(2)$; (g) $(g\circ g\circ g)'(1)$.

Problem 10 *Implicit differentiation.*

First check that the point is on the curve; then differentiate — every y you differentiate brings a y' — and solve for y'; only then substitute the point.

(a) $3x^2+y^2=7$: find y' at $(1,2)$.

(b) $x^2+xy+2y^2=28$: find y' in terms of x and y, and the equation of the tangent line at $(2,3)$.

(c) On the curve of (b), find the points at which the tangent line is horizontal.

(d) $x^2y+y^3=10$: find y' at $(1,2)$.

(e) $y+\sin y=x$: find y' in terms of y, and its value at $(0,0)$.

Problem 11 *The folium, continued.*

Lecture 10 differentiated the folium of Descartes, $x^3+y^3=9xy$, and found $y'=\dfrac{3y-x^2}{y^2-3x}$.

(a) Find the point of the folium in the first quadrant, other than the origin, at which the tangent line is horizontal. (Lecture 10 left this to the tutorial: $y'=0$ means $3y=x^2$; substitute into the equation of the curve.)

(b) Where is the tangent vertical? (Use the symmetry of the curve in the line $y=x$, or do the computation again with $y^2=3x$.)

(c) Find the slope and the equation of the tangent line at the point $\left(\frac{9}{2},\frac{9}{2}\right)$, which lies on the curve. What is special about the direction?

Problem 12 *Second derivatives by implicit differentiation.*

(Not in the lecture.) To find y'': differentiate y' again, with the product or quotient rule, and replace y' by the expression already found; use the equation of the curve to simplify.

(a) $x^2-2y^2=2$: find y' and y'' in terms of x and y, simplify y'' using the equation of the curve, and evaluate both at $(2,1)$.

(b) $y^2=x^3$: find y' and y'' at the point $(4,8)$.

Challenge *(for those who finish early, or at home)*

1. Problem 5(c) gave $p(x)=x|x|$ with $p'(x)=2|x|$. Show that p is differentiable everywhere but p' is not differentiable at 0: a function can be differentiable once and not twice.
2. Let $f(x)=\dfrac{1}{1-x}$. Compute f', f'' and f''', guess a formula for $f^{(n)}(x)$, and prove it by induction.
3. Let $f(x)=x^2\sin\dfrac{1}{x}$ for $x\neq 0$ and $f(0)=0$. Show from the definition that $f'(0)=0$ (Squeeze Theorem). Then compute $f'(x)$ for $x\neq 0$ and show that f' has no limit at 0: f is differentiable everywhere, yet f' is *not continuous* at 0.